

Review Article

The comparative efficacy and safety of sugammadex and neostigmine in reversing neuromuscular blockade in adults. A Cochrane systematic review with meta-analysis and trial sequential analysis

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Summary

We compared the efficacy and safety of sugammadex and neostigmine in reversing neuromuscular blockade in adults. Our outcomes were: recovery time from second twitch to train-of-four ratio > 0.9 ; recovery time from post-tetanic count 1–5 to train-of-four ratio > 0.9 ; and risk of composite adverse and serious adverse events. We searched for randomised clinical trials irrespective of publication status and date, blinding status, outcomes reported or language. We included 41 studies with 4206 participants. Time to reversal of neuromuscular blockade from second twitch to a train-of-four ratio > 0.9 was 2.0 min with sugammadex 2 mg.kg⁻¹ and 12.9 min with neostigmine 0.05 mg.kg⁻¹, with a mean difference (MD) (95%CI) of 10.2 (8.5–12.0) ($I^2 = 84\%$, 10 studies, $n = 835$, Grades of Recommendation, Assessment, Development and Evaluation (GRADE): moderate quality). Time to reversal of neuromuscular blockade from a post-tetanic count of 1–5 to a train-of-four ratio > 0.9 was 2.9 min with sugammadex 4 mg.kg⁻¹ and 48.8 min with neostigmine 0.07 mg.kg⁻¹, with a MD (95%CI) of 45.8 (39.4–52.2) ($I^2 = 0\%$, 2 studies, $n = 114$, GRADE: low quality). There were significantly fewer composite adverse events in the sugammadex group compared with neostigmine, with a risk ratio (95%CI) of 0.60 (0.49–0.74) ($I^2 = 40\%$, 28 studies, $n = 2298$, number needed to treat (NNT): 8, GRADE: moderate quality). Specifically, the risk of bradycardia (RR (95%CI) 0.16 (0.07–0.34), $n = 1218$, NNT: 14, GRADE: moderate quality), postoperative nausea and vomiting (RR (95%CI) 0.52 (0.28–0.97), $n = 389$, NNT: 16, GRADE: low quality) and overall signs of postoperative residual paralysis (RR (95%CI) 0.40 (0.28–0.57), $n = 1474$, NNT: 13, GRADE: moderate quality) were all reduced. There was no significant difference regarding the risk of serious adverse events (RR 0.54, 95%CI 0.13–2.25, $I^2 = 0\%$, $n = 959$, GRADE: low quality). Sugammadex reverses neuromuscular blockade more rapidly than neostigmine and is associated with fewer adverse events.

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Introduction

Neuromuscular blocking agents are used to facilitate tracheal intubation, protect the patient's vocal cords from injury, ensure patient immobility and improve surgical conditions by suppressing voluntary or reflex skeletal muscle movements [1]. Postoperative residual curarisation associated with the use of non-depolarising neuromuscular blocking agents can lead to pulmonary complications such as impaired upper airway function, increased risk of aspiration and respiratory insufficiency, and postoperative decrease in muscle strength, resulting in impairment of vision and delayed recovery and discharge time [2, 3].

Acetylcholine inhibitors such as neostigmine have been widely used to reverse the actions of non-depolarising neuromuscular blocking agents by competitive antagonism [4, 5]. Their indirect mechanism of action means that, the reversal can be limited and unpredictable, resulting in potential risk of post-operative residual curarisation. Furthermore, undesirable autonomic responses such as bradycardia, hypotension, bronchoconstriction, airway secretions and increased gastrointestinal motility can occur, due to stimulation of muscarinic cholinergic synapses. Consequently, antimuscarinic drugs such as glycopyrrolate or atropine are usually given with neostigmine, and may in turn cause tachycardia, dry mouth, urinary retention and bronchodilatation [4].

Sugammadex is a synthetically modified gamma-cyclodextrin with a hydrophilic exterior and a hydrophobic core, specifically designed to encapsulate rocuronium and reverse rocuronium-induced neuromuscular blockade [6–8]. Due to this direct mechanism of action, sugammadex is associated with fast and predictable reversal of any degree of block. The undesirable side-effects of neostigmine and antimuscarinic drugs are also avoided. There are also potential benefits in terms of reduced incidence of postoperative residual curarisation, reduced duration of anaesthesia, higher flow of patients through the operating theatre and more efficient use of healthcare resources.

The aim of this review was to assess the efficacy and safety of sugammadex compared with neostigmine in adult patients.

Methods

This systematic review was carried out in accordance with Cochrane Collaboration methodology, PRISMA and GRADE guidelines [9–12]. This paper is a revised update of a previously published Cochrane review [13].

We included randomised clinical trials irrespective of publication status, date of publication, blinding status, outcomes reported or language of the report. We planned to contact trial investigators and authors to ask for relevant data. We included adults (> 18 years of age), classified as ASA physical status 1–4, who received non-depolarising neuromuscular blocking agents for an elective in-patient or day-surgical procedure. We included all trials comparing sugammadex with neostigmine in adults receiving non-depolarising neuromuscular blocking agents. We included any dose of sugammadex and neostigmine and any time-point of administration of study drug.

Our outcomes were: recovery time from moderate neuromuscular blockade from re-appearance of second twitch to train-of-four ratio > 0.9; recovery time from deep neuromuscular blockade from re-appearance of post-tetanic count 1–5 to train-of-four ratio > 0.9; and risk of adverse and serious adverse events.

For the first outcome recovery time from second twitch to train-of-four ratio > 0.9, we compared sugammadex 2 mg.kg⁻¹ with neostigmine 0.05 mg.kg⁻¹. For the outcome recovery time from post-tetanic count 1–5 to train-of-four ratio > 0.9, we compared sugammadex 4 mg.kg⁻¹ with neostigmine 0.07 mg.kg⁻¹. Recovery time was measured in minutes from administration of the study drug to train-of-four ratio > 0.9. For the risk of adverse and serious adverse events we compared any administered dose of sugammadex and neostigmine, regardless of time of administration. Adverse events and serious adverse events were defined by study authors and observed and assessed by safety outcome assessors in the operating theatre, postanesthetic care unit or up to seven days after surgery, depending on each study. The risk of adverse events was measured as number of adverse events per all patients. Only adverse events that were possibly, probably or definitely related to the study drug were included in the risk assessments.

In this updated review, we searched the Cochrane Central Register of Controlled Trials (CENTRAL; 2016, Issue 4); MEDLINE (Ovid SP, 1950 to 2 May 2016) and Embase (Ovid SP, 1980 to 2 May 2016). Furthermore, we searched for ongoing clinical trials and unpublished trials on the following Internet sites: www.controlled-trials.com, clinicaltrials.gov and www.centerwatch.com. We performed a search update on 10 May 2017. For specific information regarding our search strategies, see Supporting Information, Table S1.

All trials were evaluated for major potential sources of bias using the Cochrane risk of bias tool (random sequence generation; allocation concealment; blinding of participants; blinding of personnel; blinding of primary outcome assessor; blinding of secondary outcome assessor; incomplete outcome data; selective reporting; funding bias; and other bias). We assessed each domain separately and in total, grading each domain 'high risk', 'low risk' or 'unclear risk' of bias [9, 14]. For more detailed information on the Cochrane risk of bias tool, see Supporting Information, methods section.

We used Review Manager (RevMan) software, version 5.3 (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, the Netherlands) and calculated mean differences (MD) with 95%CI for continuous outcomes and risk ratio (RR) with 95%CI for dichotomous variables. We used the Chi-square test to obtain an indication of heterogeneity between trials, with a $p \leq 0.1$ considered significant. We quantified the degree of heterogeneity observed in the results by using the I^2 statistic.

We used the principles of Grades of Recommendation, Assessment, Development and Evaluation (GRADE) approach to provide an overall assessment of evidence relating to our outcomes, evaluating within-study risk of bias, inconsistency, indirectness, imprecision, risk of publication bias and other factors. The GRADE assessment resulted in one of four levels of 'quality': high, moderate, low or very low, and these expressed our confidence in the estimate of effect [11, 15].

We used trial sequential analysis to examine the required information size. Trial sequential analysis is a methodology that combines the total accrued sample size of all included trials relative to the required

information size with an adjusted threshold for statistical significance [16–19]. For more detailed information on this technique, see the Supporting Information, methods section.

Results

Full results are available in the Cochrane version [61].

Search results are displayed in a PRISMA flow chart (Fig. 1). Forty-one trials [20–60] with a total of 4206 participants met our inclusion criteria of which 31 trials [20–34, 36–44, 46, 48, 53, 55, 58–60] with 2559 participants were eligible for meta-analysis. The remaining ten trials [35, 43, 45, 47, 50–52, 54, 56, 57] with 1647 participants were included in the qualitative analysis. Furthermore, 20 trials are ongoing and three are awaiting classification. Of the 41 trials, 29 were published as full-text papers [20, 22–27, 29, 30, 33–40, 42, 46–50, 53, 54, 57–60] while 12 were only available as meeting abstracts [21, 28, 31, 32, 41, 43–45, 51, 52, 55, 56]. All of the included trials were published in English with the exception of one article published in Turkish [40]. All 41 authors were contacted for missing information; 12 of them (29%) replied and provided supplementary data. The trials were conducted in various centres across Europe, Unites States and Asia, 30 of which were single-centre studies and 11 were multicentre studies. The sample size of the

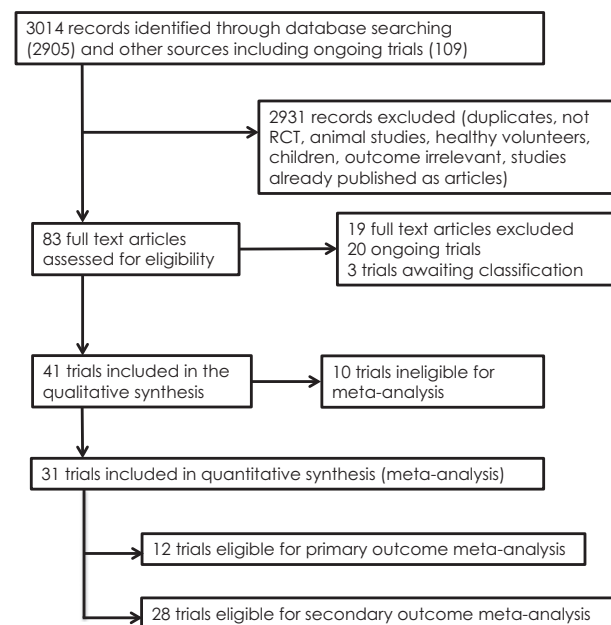


Figure 1 Flow diagram of search and trial selection.

included trials ranged from 22 to 1198 adults. The participants in the included trials underwent various elective surgical procedures under different types of general anaesthesia, using various types of neuromuscular blocking agents and doses of reversal study drugs. Additionally, some trials only included specific types of patients. For a detailed description on these points, see Supporting Information, results section. None of the included trials had a low risk of bias across all domains; in particular, performance bias and funding bias were high overall. The risk of bias assessment is provided in the Supporting Information, Figure S1.

All trials assessed the train-of-four ratio using acceleromyography on the same monitoring site (ulnar nerve and adductor pollicis muscle). Ten trials [22, 26, 28, 29, 31, 32, 34, 42, 58, 59] were included in the meta-analysis of recovery time from second twitch to train-of-four ratio > 0.9 . Sugammadex 2 mg.kg⁻¹ reversed moderate neuromuscular blockade from second twitch to train-of-four ratio > 0.9 in 2.0 min in comparison with 12.9 min for neostigmine 0.05 mg.kg⁻¹ (MD (95%CI) 10.2 (8.5–12.0), $I^2 = 84\%$, $n = 835$, random-effects model, GRADE quality of evidence: moderate, Fig. 2). The GRADE quality of evidence was downgraded one level due to high risk of bias.

The trial sequential analysis indicates that with a required information size of 106, firm evidence is in place in favour of sugammadex (Fig. 3). However, none of the included trials had a low risk of bias, and since trial sequential analysis is ideally designed for

trials with low risk of bias and is unable to adjust for risk of bias, the precision of our findings has to be downgraded. Furthermore, there was a high degree of diversity and heterogeneity, which once again raises questions about the reliability of the calculated required information size.

Two trials [24, 36] were combined in the meta-analysis of recovery time from post-tetanic count 1–5 to train-of-four ratio > 0.9 . Sugammadex 4 mg.kg⁻¹ reversed deep neuromuscular blockade from post-tetanic count 1–5 to train-of-four ratio > 0.9 in 2.9 min compared with 48.8 min for neostigmine 0.07 mg.kg⁻¹ (MD (95%CI) 45.8 (39.4–52.2), $I^2 = 0\%$, $n = 114$, random-effects model, GRADE quality of evidence: low). The GRADE quality of evidence was downgraded one level due to high risk of bias and one level due to imprecision. Trial sequential analysis was not performed.

Twenty-eight trials [20–27, 29, 30, 33, 34, 36–43, 46, 48, 49, 53, 54, 58–60] investigated the risk of adverse events possibly, probably or definitely related to the study drugs. Meta-analysis indicated significantly fewer composite adverse events in the sugammadex group compared with the neostigmine group, RR (95%CI) 0.60 (0.49–0.74) ($I^2 = 40\%$, $n = 2298$, random-effects model, GRADE quality of data: moderate, Fig. 4). The GRADE quality of evidence was downgraded one level due to high risk of bias. The risk of composite adverse events was 283/1000 in the neostigmine group and 159/1000 in the sugammadex group, resulting in a number needed to treat (NNT) of 8 in order to avoid an adverse event.

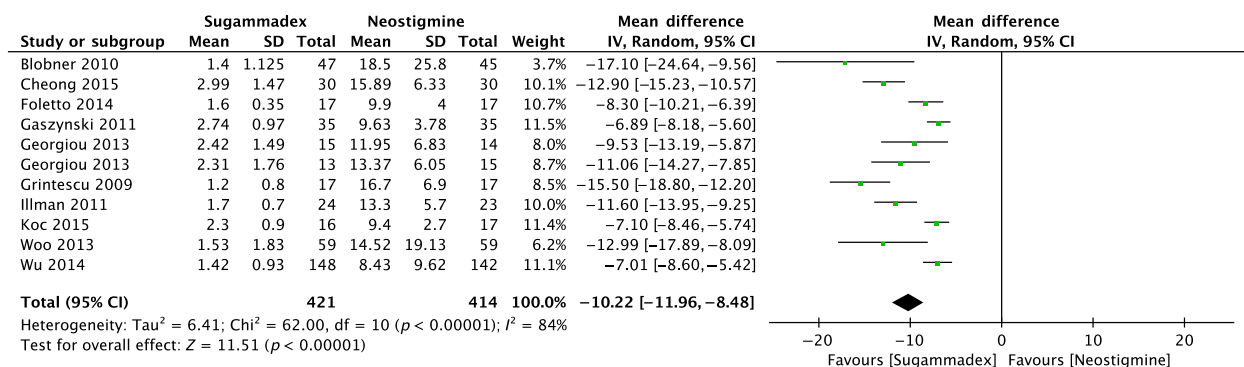


Figure 2 Forest plot of recovery time from second twitch to train-of-four ratio > 0.9 ; sugammadex 2.0 mg.kg⁻¹ vs. neostigmine 0.05 mg.kg⁻¹. IV, inverse variance.

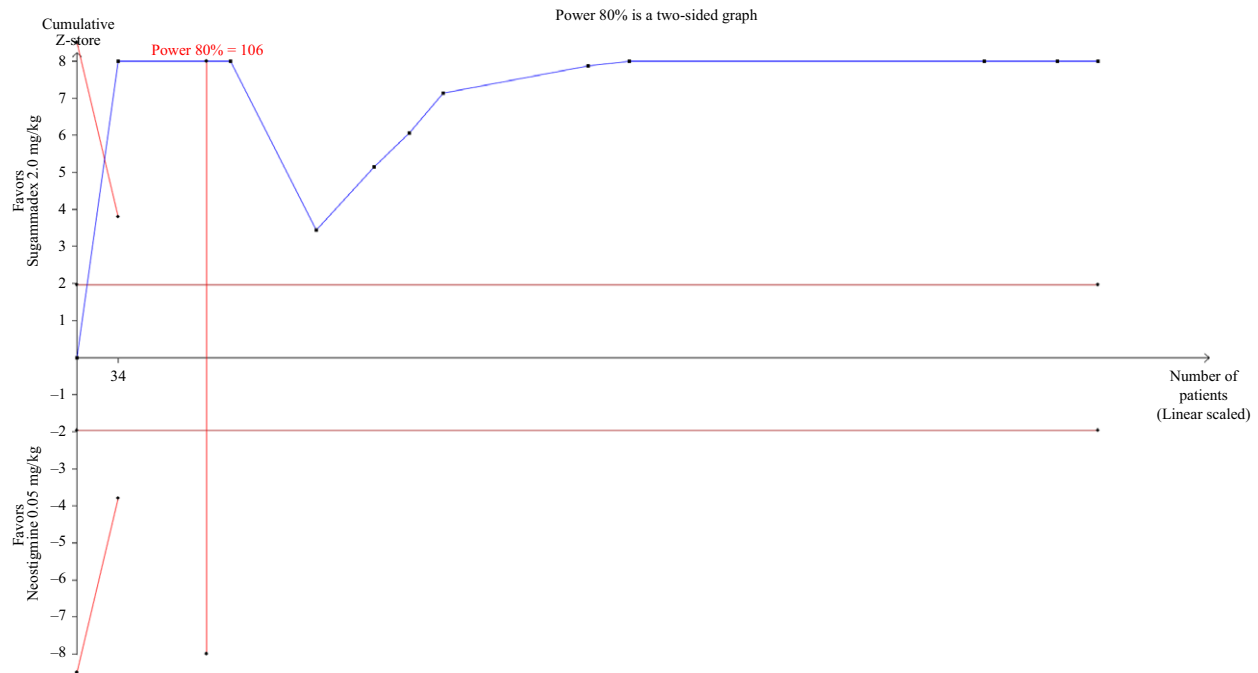


Figure 3 Trial sequential analysis of recovery time from second twitch to train-of-four ratio > 0.9 min; sugammadex 2.0 mg.kg⁻¹ vs. neostigmine 0.05 mg.kg⁻¹.

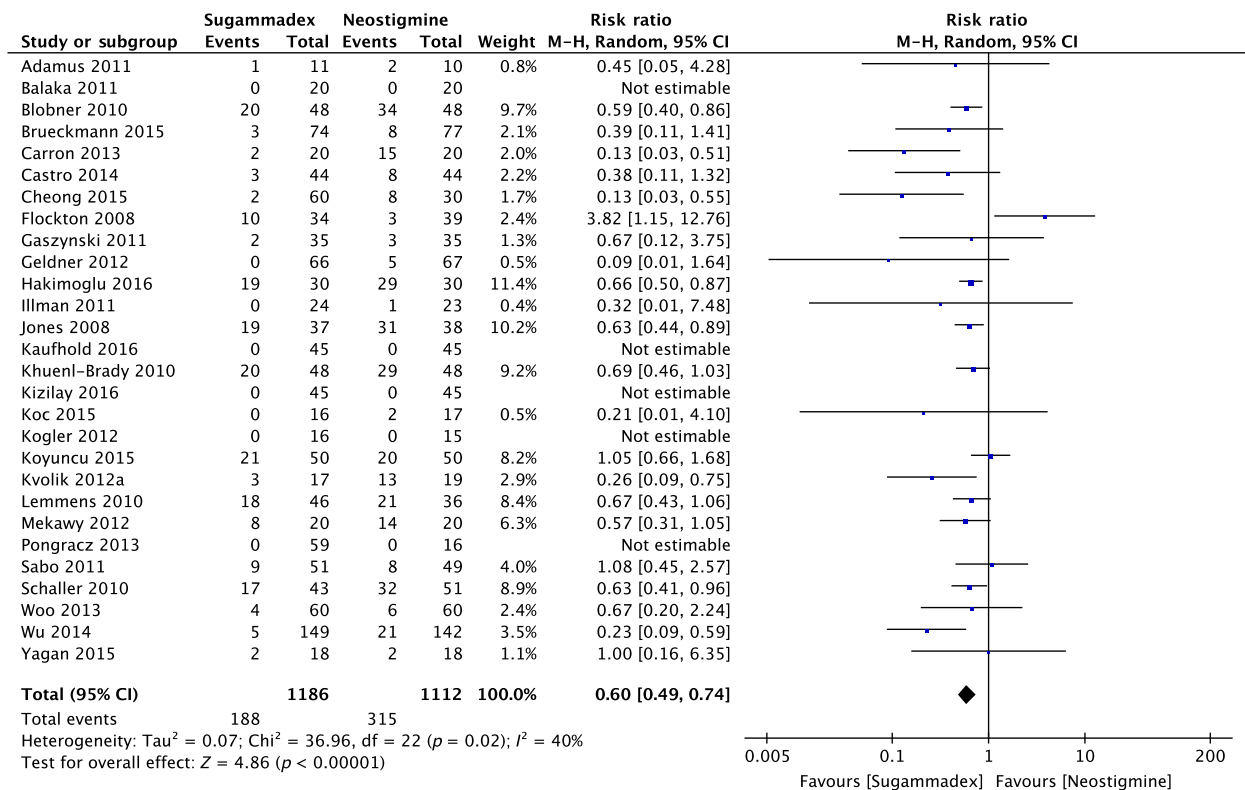


Figure 4 Forest plot of risk of adverse events; sugammadex (any dose) vs. neostigmine (any dose). M-H, Mantel-Haenszel.

A trial sequential analysis of drug related risk of adverse events when comparing neostigmine with sugammadex at any dose resulted in a required information size of 502, indicating firm evidence in favour of sugammadex, as 2298 participants were included in our analysis. Despite the fact that the cumulative Z-curve does not cross the monitoring boundary directly, it is hard to imagine future trials radically changing the overall picture of this analysis. However, none of the included trials were at low risk of bias, which downgrades the reliability of our finding.

When looking at specific adverse events, a significantly lower risk of the following adverse events in the sugammadex group was detected when compared with neostigmine: bradycardia, RR (95%CI) 0.16 (0.07–0.34) ($I^2 = 0\%$, $n = 1218$, random-effects model, NNT: 14, GRADE quality of data: moderate, downgraded one level due to high risk of bias); and postoperative nausea and vomiting, RR (95%CI) 0.52 (0.28–0.97) ($I^2 = 0\%$, $n = 389$, random effects model, NNT: 16, GRADE quality of data: low, downgraded one level due to high risk of bias and one level due to imprecision).

Sub-group analysis of our data revealed no significant sub-group difference in the RR of composite adverse events when taking into account the different doses of sugammadex and neostigmine as well as the type of general anaesthesia (total intravenous anaesthesia vs. volatile anaesthetics). Sub-group analysis of the

bradycardia data revealed no significant sub-group difference in the RR of composite adverse events when atropine was compared with glycopyrrolate. Sub-group analysis of the postoperative nausea and vomiting data revealed no significant sub-group difference in the RR of composite adverse events when taking into account the type of general anaesthesia (total intravenous anaesthesia vs. volatile anaesthetics).

We chose the following parameters as overall signs of postoperative residual paralysis: inability to perform 5-s head lift test and general muscle weakness after extubation and at post-anaesthesia care unit discharge; amblyopia, subjective complaint of weakness, oxygen desaturation $< 90\%$; transitory oxygen supplementation; respiratory distress; respiratory depression; postoperative respiratory complications; moderate dyspnoea; pneumonia; acute lung failure; or if the authors specifically reported symptoms of residual neuromuscular blockade or recurrence of neuromuscular blockade. Fifteen studies reported one or more of these adverse events [21–24, 27, 30, 36, 38, 42, 44, 46, 48, 54, 58, 59]. Meta-analysis of the results showed a significantly reduced risk of overall signs of postoperative residual paralysis, RR (95%CI) 0.40 (0.28–0.57) ($I^2 = 0\%$, $n = 1474$, random-effects model, NNT: 13, GRADE quality of evidence: moderate, Fig. 5) in the sugammadex group. GRADE quality of evidence was downgraded one level due to high risk of bias.

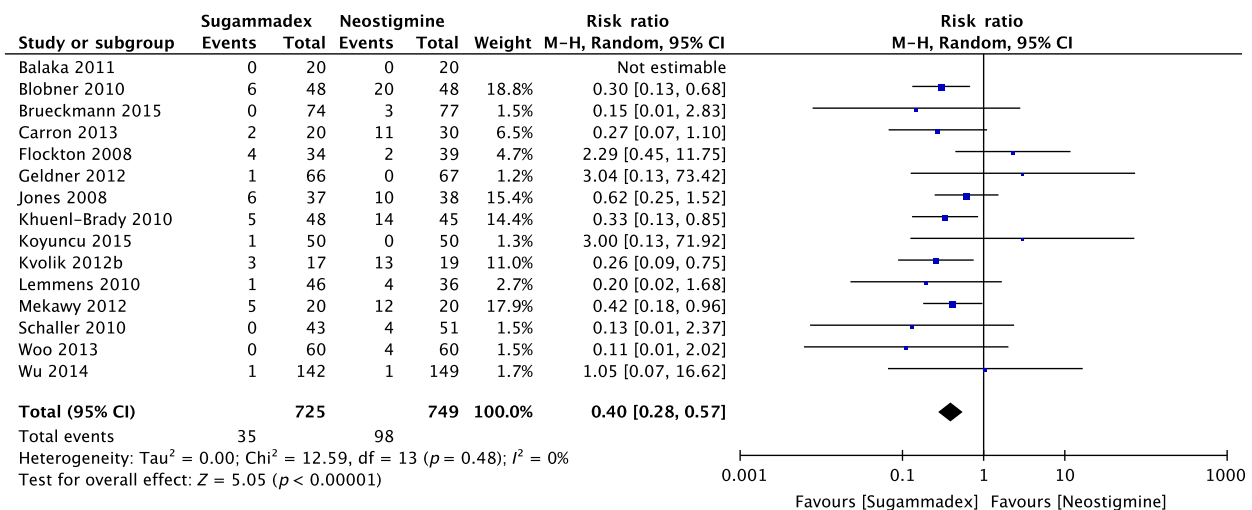


Figure 5 Forest plot of risk of overall signs of postoperative residual paralysis; sugammadex (any dose) vs. neostigmine (any dose). M-H, Mantel-Haenszel.

A trial sequential analysis of the risk of overall signs of postoperative residual paralysis when comparing any dose of sugammadex with neostigmine resulted in a required information size of 424 participants, indicating firm evidence in favour of sugammadex. None of the included trials had low risk of bias and this equally diminishes the reliability and precision of our estimates.

The effects of sugammadex and neostigmine on intra-ocular pressure [33, 60]; haemodynamics [39]; bleeding events [48, 50, 57]; renal function [35]; gastric emptying [56], thyroid function [43]; cognitive function [52]; postoperative nausea and vomiting [24, 51, 57]; and pain [47, 57] were reported in data format ineligible for meta-analysis.

Fourteen trials [20, 22, 23, 27, 30, 33, 36, 37, 42, 46, 54, 58, 59] reported serious adverse events possibly, probably or definitely related to the study drug. Meta-analysis of the results showed no significant difference between sugammadex and neostigmine regarding composite serious adverse events, RR (95%CI) 0.54 (0.13–2.25) ($I^2 = 0\%$, $n = 959$, random-effects model, GRADE quality of evidence: low). GRADE quality of evidence was downgraded one level due to high risk of bias and one level due to imprecision. Trial sequential analysis was not performed.

Clearly reported drug-related serious adverse events included: one case each of acute myocardial infarction, pneumonia and inadequate reversal of neuromuscular blockade in the neostigmine group [23]; one case of acute lung failure in the neostigmine group [54]; one case of postoperative upper abdominal pain in the neostigmine group [30]; one case of post-procedural haemorrhage in the sugammadex group [23]; and one case of respiratory depression in the sugammadex group [42].

Discussion

Our systematic review provides a robust assessment of efficacy and safety as it includes a large number of trials with a consistent direction of effect. It also provides additional confirmation through various exploratory analyses which all favour sugammadex for all outcomes analysed. However, there are several potential limitations as our findings and interpretations are limited by the quality and quantity of available evidence in the included trials. All trials had at least one domain at high or unclear risk of bias and the risk of over- or

underestimation of the true intervention effect was considered a serious limitation. In particular, performance bias and funding bias were high overall. Furthermore, the risk of bias of the included trials was mainly assessed using published data, which ultimately may not reflect the truth. All trial authors were contacted but only one-third responded and provided further information. Lack of reporting may have affected our ability to correctly judge of risk of bias in either direction. None of the studies were judged as having a low risk of bias. Application of the GRADE approach enabled us to incorporate risk of bias, directness of evidence, heterogeneity, precision of effect estimate and risk of publication bias. According to the GRADE system, the quality of our findings ranked low to moderate across different outcomes. The main limiting factors that accounted for a decrease in overall quality were high risk of bias and imprecision. Application of trial sequential analysis indicated that, at this stage, sugammadex appears superior to neostigmine. However, as none of the included trials were at low risk of bias, and since trial sequential analysis is unable to adjust for the risk of bias, the low risk of bias adjusted information size has not been calculated, which ultimately affects the reliability of our findings.

Moderate neuromuscular blockade usually provides sufficient relaxation for most surgical procedures. Deep neuromuscular blockade can be required in precision procedures where unexpected movements can be deleterious (vocal cord and eye laser surgery, neurosurgery, neuroradiological embolisation, robot-guided ablation) or in procedures where it is necessary to relax muscles as much as possible (orthopaedic fracture repositioning, dislocation reduction, laparotomy, laparoscopy [62–64]. If deep blockade is required until the 'last stitch' or if surgery is terminated unexpectedly early, sugammadex enables the anaesthesiologist to reverse blockade quickly and reliably. Administration of neostigmine for reversal of deep block in the absence of any signs of neuromuscular recovery is not recommended due to the 'ceiling effect' seen when maximal acetylcholine concentration is unable to adequately compete with muscle relaxant [3]. Even though this is considered an unlicensed indication for neostigmine use [65], our search identified three trials using this technique (all included in our review) [24, 36, 46].

The studied drug doses for reversal of moderate and deep neuromuscular blockade were selected as they are clinically most frequently used, also reflected in the notion that most of the included studies in our review used these doses.

Monitoring and reporting of adverse events during a clinical trial is a cumbersome and complex task involving many assumptions and choices, such as adequate blinding of patients and investigators, distinction between adverse and serious adverse events, attribution of adverse events to study drugs, reporting by patients and finally consistent and transparent monitoring, coding and reporting by investigators. The trials included in this review defined, monitored and reported adverse events in many different ways. Some trials [22, 36, 46] coded all adverse and serious adverse events in a systematic way using Medical Dictionary for Regulatory Activities (MedDRA). Other trials reported symptoms related to study drug administration without necessarily defining them as adverse events [20, 48], an issue most often seen in meeting abstracts [21, 31, 44] probably due to word count restriction. Some of the included trials specifically addressed causality between adverse events and study drugs by presenting both adverse events observed 'regardless of relation to study drug' and adverse events 'possibly, probably or definitely related to study drug' [22, 36, 46, 58] while other trials did not specifically address this issue [20, 25, 60]. Smaller trials with few observed adverse events usually presented all observed adverse events [20, 21, 40, 60], while bigger trials presented most frequently occurring adverse events [23, 36, 46, 58]. Additionally, some trials used blinded safety outcome assessors [22–24, 27, 58] in contrast with others [32, 39]. Last but not least, very few studies were designed and powered to address safety as primary outcome [23, 50]. Furthermore, we also decided to include reported symptoms related to drug administration when they were not specifically labelled as adverse events in order not to potentially dismiss good quality data due to lack of correct phrasing. Finally, we performed meta-analyses of both specific and composite adverse events [66]. Readers of this review need therefore to be aware of all these issues in order to appraise our data critically.

We pooled all doses of sugammadex and neostigmine in a single analysis as we did not expect any dose-

dependence of adverse events. Furthermore, many of the included studies used different drug doses, as described in the Supporting Information, results section.

Bradycardia secondary to neostigmine is seen due to stimulation of muscarinic cholinergic receptors throughout the autonomic nervous system and usually requires administration of antimuscarinic drugs [4, 5]. Our results indicate that the direct mechanism of action of sugammadex bypasses this issue. However, most studies included in this analysis did not provide a clear definition of bradycardia nor time frame of its occurrence. Furthermore, atropine induces its vagolytic effect more rapidly than glycopyrrolate does [67] but sub-group analysis revealed no difference between groups.

Reports on the effect of neostigmine on postoperative nausea and vomiting are conflicting because neostigmine has been implicated as its cause [68], described as having anti-emetic properties [69] and having no effect [70]. Only six trials included in our meta-analysis [20, 25, 26, 33, 54, 60] specifically reported postoperative nausea and vomiting. Other trials differentiated between nausea and vomiting [22, 36, 38, 40], reported only nausea [27, 34, 46, 53, 58] or differentiated between nausea and post-procedural nausea [36, 38]. Additionally, the wide confidence interval and few included patients mean that our results should be taken with caution.

Postoperative residual paralysis can be difficult to assess [2, 3]. Satisfactory recovery from neuromuscular block and clinical absence of residual curarisation has first occurred when the train-of-four ratio is > 0.9 [71]. However, some patients may exhibit obvious weakness despite achieving train-of-four ratios > 0.9 , whereas complete recovery of muscle strength may be observed in patients with train-of-four ratio ratios < 0.9 . Therefore, a precise definition of residual block requires not only the measurement of train-of-four ratios using objective neuromuscular monitoring devices but also careful clinical assessment of each patient for adverse effects potentially attributable to the use of neuromuscular blocking agents [2]. Furthermore, clinical tests such as head-lift, hand-grip and leg-lift have low sensitivity and specificity and should therefore always be supplemented with monitoring of neuromuscular blockade [72, 73]. However, monitoring of neuromuscular blockade remains relatively rarely

used due to ignorance of adverse events or non-availability monitors [3, 74, 75]. Moreover, there are other causes of postoperative critical respiratory events such as lingering effects of opioids, age, emergency surgery, long surgery duration, abdominal surgery, vascular surgery, and obesity [3]. In our review, we chose to include clinical tests and diverse adverse events described in the results section as overall signs of postoperative residual paralysis.

Our overall findings on adverse events are in line with findings of a recent systematic review comparing sugammadex and neostigmine [76], whose primary outcome was rate of postoperative residual paralysis and secondary outcomes were rates of drug-related adverse events, including postoperative nausea and vomiting. The review included 17 RCTs with 1553 participants and reported that sugammadex reduced drug-related adverse events, all signs of residual postoperative paralysis and minor respiratory events but found no difference in critical respiratory events, rate of postoperative nausea or vomiting.

In conclusion, sugammadex reverses neuromuscular blockade faster than neostigmine regardless of its depth and is associated with fewer adverse events.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Data S1. Methods.

Figure S1. Review author's judgements about risk of bias item for each included study. A green symbol denotes low risk of bias for the domain in question, yellow and red symbols denote unclear and high risk of bias, respectively.

Table S1. Search strategy for databases.